

Mobile Application Development with Flutter

Lecture 9: Lists in Dart

What is Collection?

- A **collection** is a data structure that allows us to store multiple values inside a single variable
- We'll discuss three types of collections:
 - List, Set, Map

Collection	Ordered	Allows Duplicates	Key-Value Structure
List	Yes	Yes	No
Set	No	No	No
Map	No	No (keys)	Yes

List in Dart

- A **List** is an ordered collection of elements
- List in dart is similar to **Arrays** in other programming languages
- Elements are accessed using an index
- Index starts from 0
- Duplicate values are allowed
- Demonstrate simple and mixed lists

List Operations

- Accessing items of list
- Get index by value
- Find length of list
- Modifying lists
- Mutable and immutable list

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List Properties

- First
- Last
- isEmpty
- isEmpty
- Length
- Reversed

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Adding Item to List

- `add()`
- `addAll()`
- `insert()`
- `insertAll()`
- `replaceRange()`

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Removing List Elements

- `remove()`
- `removeAt()`
- `removeLast()`
- `removeRange()`

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Misc Operations on List

- Contains
- Clear list
- Looping through a list
 - for, for-in, for-each, for each with arrow
- Multiply all values by 2 using map function
- Combine two or more lists using spread syntax
- Filter list using where method